SET NO: 19

Survey Responses Analysis

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Course: DSA06012 Data Handling and Visualization

Question 1

Create a grouped bar chart to visualize the distribution of answers for Question 1. Label the chart elements.

AIM

To visualise the distribution of answers to Question 1 using a grouped bar chart in R.

PROCEDURE

1. Create the survey data frame (or import it with read.csv()).
2. Load ggplot2 and plot geom_bar() with position = "dodge".
3. Add the title, axis labels and legend.

CODE

```
# Dataset
library(ggplot2); library(dplyr); library(tidyr); library(gridExtra)
survey <- data.frame(
  Survey_ID = c(1, 2, 3),
  Question1 = c("A", "B", "C"),
  Question2 = c("B", "A", "A"),
  Question3 = c("C", "D", "B"))
# (or) survey <- read.csv("Survey_Responses.csv")

# Grouped bar chart - Question 1
p1 <- ggplot(survey, aes(x = Question1, fill = factor(Survey_ID))) +
  geom_bar(position = "dodge", width = 0.6) + labs(title =
"Question 1 - Distribution of Answers", x = "Answer (Question
1)", y = "Count", fill = "Survey_ID") print(p1)
```

OUTPUT

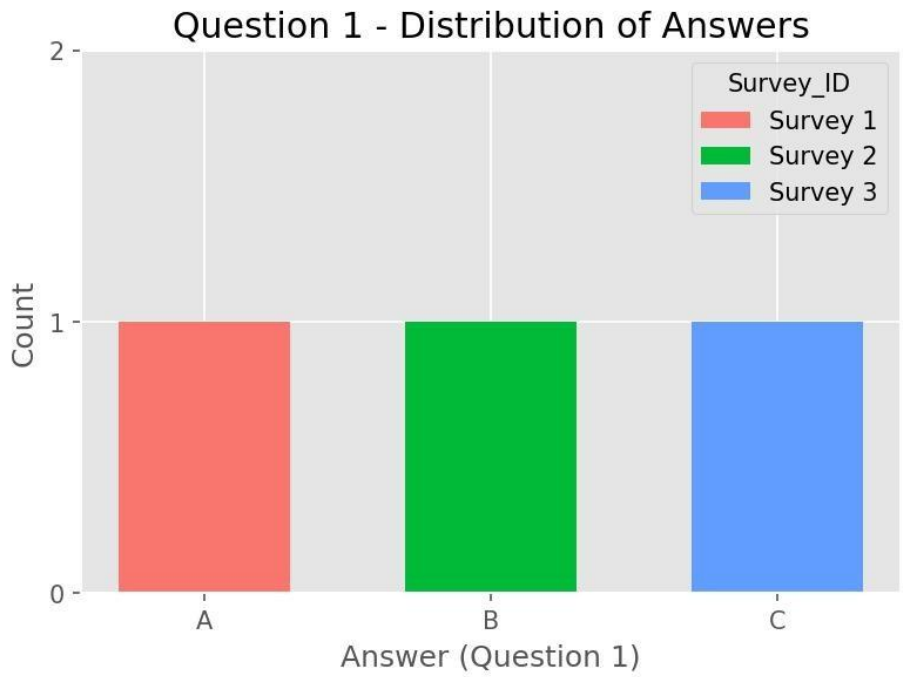


Figure 19.1: Grouped bar chart - Question 1

RESULT

The grouped bar chart shows that Question 1 received one response for each option A, B and C, so the answers are evenly distributed.

Question 2

Generate a stacked bar chart to represent the overall distribution of responses for all three questions.

AIM

To represent the overall distribution of responses for all three questions using a stacked bar chart.

PROCEDURE

1. Reshape the data into long format with pivot_longer().
2. Plot geom_bar(position = "stack") with Question on the x-axis and Answer as fill.
3. Label the title, axes and legend.

CODE

```
# Reshape to long format
long <- pivot_longer(survey, cols = starts_with("Question"),
                     names_to = "Question", values_to = "Answer")

# Stacked bar chart - all three questions
p2 <- ggplot(long, aes(x = Question, fill = Answer)) +
  geom_bar(position = "stack", width = 0.5) +
  labs(title = "Overall Distribution of Responses (Q1-Q3)", x =
"Question", y = "Number of Responses", fill = "Answer") print(p2)
```

OUTPUT

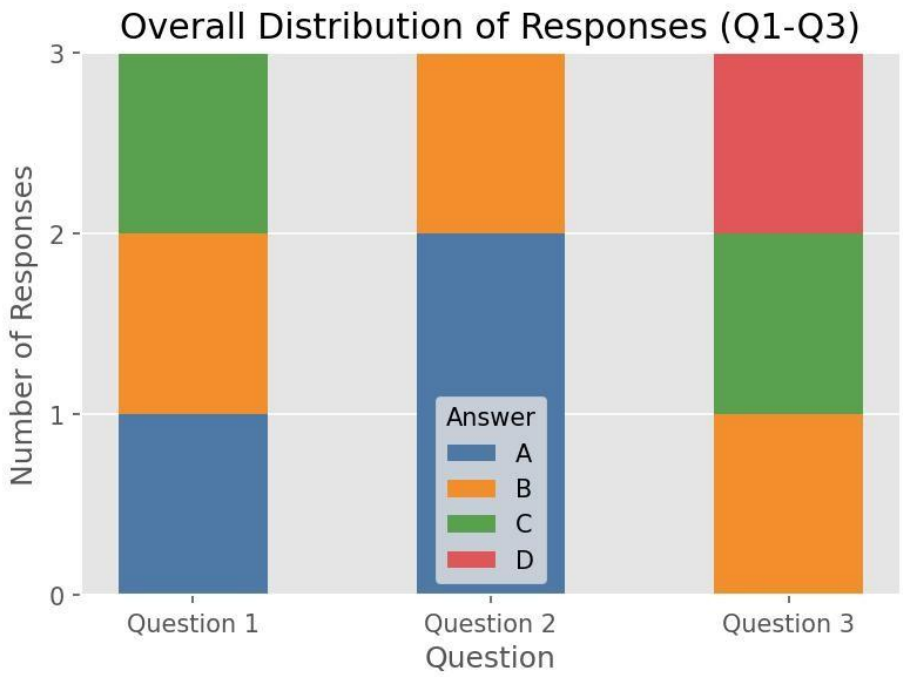


Figure 19.2: Stacked bar chart - Questions 1 to 3

RESULT

Option A is the most frequent answer for Question 2 (2 of 3 responses). Questions 1 and 3 have responses spread across different options (A, B, C and B, C, D).

Question 3

Build a table to show the survey response data for each survey. Label the table elements.

AIM

To display the survey response data of each survey in a labelled table.

PROCEDURE

1. Print the data frame using print().
2. Convert it using tableGrob() from gridExtra.
3. Display the table with a title.

CODE

```
# Table of survey responses
print(survey)
tbl <- tableGrob(survey, rows = NULL)
grid.newpage()
grid.arrange(tbl, top = "Survey Response Data")
```

OUTPUT

Survey Response Data

Survey ID	Question 1	Question 2	Question 3
1	A	B	C
2	B	A	D
3	C	A	B

Table 19.1: Survey response data

RESULT

The table lists the answers given by each of the three surveys for Questions 1, 2 and 3.

Question 4

Develop a dashboard combining the grouped bar chart, stacked bar chart, and the table for interactive exploration of survey responses.

AIM

To combine the grouped bar chart, stacked bar chart and table into one dashboard for exploring the survey responses.

PROCEDURE

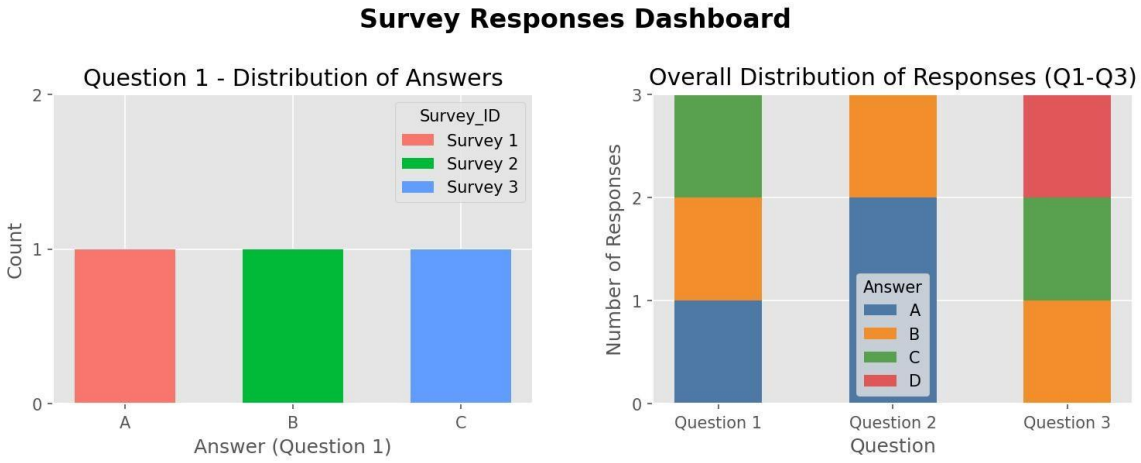
1. Reuse p1, p2 and the table tbl from the previous questions.
2. Arrange them with grid.arrange() under a common title.
3. Use plotly (ggplotly) and DT for interactive exploration.

CODE

```
# Dashboard: grouped bar + stacked bar + table
dash <- grid.arrange(
arrangeGrob(p1, p2, ncol = 2), tbl, nrow = 2,
heights = c(2, 1), top = "Survey Responses
Dashboard")

# Interactive exploration (hover, zoom, filter)
library(plotly) ggplotly(p1) ggplotly(p2)
library(DT)
datatable(survey, filter = "top")
```

OUTPUT



Survey ID	Question 1	Question 2	Question 3
1	A	B	C
2	B	A	D
3	C	A	B

Figure 19.3: Survey responses dashboard

RESULT

The dashboard displays the grouped bar chart, stacked bar chart and the response table together, and the plotly and DT versions allow interactive exploration of the survey responses.